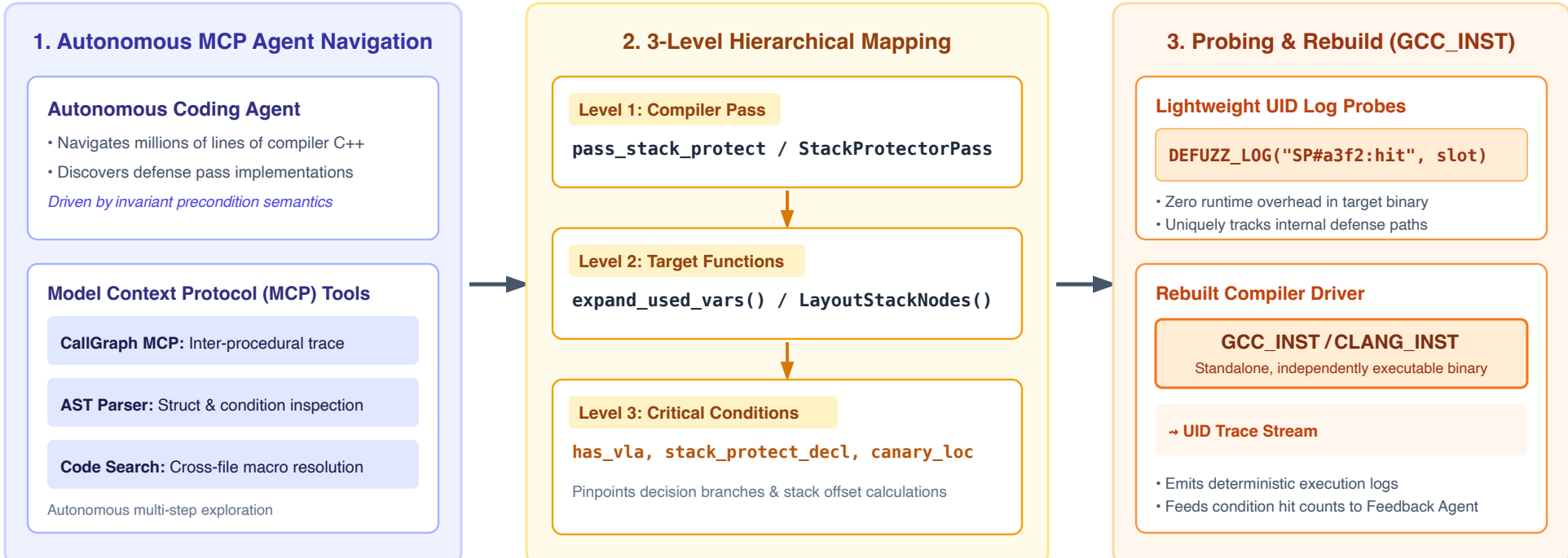

1. Autonomous MCP Agent Navigation
Autonomous Coding Agent
• Navigates millions of lines of compiler C++
• Discovers defense pass implementations
Driven by invariant precondition semantics
Model Context Protocol (MCP) Tools
CallGraph MCP: Inter-procedural trace
AST Parser: Struct & condition inspection
Code Search: Cross-file macro resolution
Autonomous multi-step exploration
2. 3-Level Hierarchical Mapping
Level 1: Compiler Pass
pass_stack_protect / StackProtectorPass
Level 2: Target Functions
expand_used_vars() / LayoutStackNodes()
Level 3: Critical Conditions
has_vla, stack_protect_decl, canary_loc
Pinpoints decision branches & stack offset calculations
3. Probing & Rebuild (GCC_INST)
Lightweight UID Log Probes
DEFUZZ_LOG("SP#a3f2:hit", slot)
• Zero runtime overhead in target binary
• Uniquely tracks internal defense paths
Rebuilt Compiler Driver
GCC_INST / CLANG_INST
Standalone, independently executable binary
⇝ UID Trace Stream
• Emits deterministic execution logs
• Feeds condition hit counts to Feedback Agent